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RESEARCH ARTICLE

Enhancing Early Detection of Diabetic Retinopathy Through the Integration of Deep Learning Models and Explainable Artificial Intelligence

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ABSTRACT Humans can carry various diseases, some of which are poorly understood and lack comprehensive solutions. Such a disease can exists in human eye that can affect one or both eyes is diabetic retinopathy (DR) which can impair function, vision, and eventually result in permanent blindness. It is one of those complex complexities. Therefore, early detection of DR can significantly reduce the risk of vision impairment by appropriate treatment and necessary precautions. The primary aim of this study is to leverage cutting-edge models trained on diverse image datasets and propose a CNN model that demonstrates comparable performance. Specifically, we employ transfer learning models such as DenseNet121, Xception, Resnet50, VGG16, VGG19, and InceptionV3, and machine learning models such as SVM, and neural network models like (RNN) for binary and multi-class classification. It has been shown that the proposed approach of multi-label classification with softmax functions and categorical cross-entropy works more effectively, yielding perfect accuracy, precision, and recall values. In particular, Xception achieved an impressive 82% accuracy among all the transfer learning models, setting a new benchmark for the dataset used. However, our proposed CNN model shows superior performance, achieving an accuracy of 95.27% on this dataset, surpassing the state-of-the-art Xception model. Moreover, for single-label (binary classifications), our proposed model achieved perfect accuracy as well. Through exploration of these advances, our objective is to provide a comprehensive overview of the leading methods for the early detection of DR. The aim is to discuss the challenges associated with these methods and highlight potential enhancements. In essence, this paper provides a high-level perspective on the integration of deep learning techniques and machine learning models, coupled with explainable artificial intelligence (XAI) and gradient-weighted class activation mapping (Grad-CAM). We present insights into their respective accuracy and the challenges they face. We anticipate that these insights will prove valuable to researchers and practitioners in the field. Our ambition is that this in-depth study and comparison of models will inform

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and inspire future research endeavors, ultimately leading to enhanced disease detection in medical imaging and thereby assisting healthcare professionals.

INDEX TERMS Diabetic retinopathy, transfer learning, CNN, Xception, inception, Grad-CAM.

I. INTRODUCTION

Diabetic retinopathy (DR) is a widely prevalent ailment, and regrettably, no definitive cure exists; however, various approaches have been devised to ameliorate its symptoms. The alarming ubiquity of DR is a matter of profound concern, compounded by the growing consensus among the scientific community that individuals living with diabetes are at a substantial and escalating risk of developing this condition. According to [1], this condition accounts for 4.8% of the 37 billion global cases of blindness. Diagnosing DR relies on a complex set of limited features and locations in the retinal images. According to estimates from the World Health Organization (WHO), 422 million people worldwide suffer from diabetes, and the prevalence of diabetes has increased from 4.7% to 8.5% globally, as reported in [2]. It is projected that by 2030, there will be 191.0 million people with diabetes worldwide, as described in [3]. Currently, DR and diabetic macular edema (DME) have respective prevalence rates of 35% and 7%, according to [4]. Diagnosing diabetic retinopathy in its early stages is exceptionally challenging because microaneurysms, capillary outflow, retinal hemorrhages, and blood vessel breaches are typically concealed at that point. To reduce the burden on ophthalmologists, researchers have developed a computerized technique that can detect and classify anomalies in the retina based on their severity. Nevertheless, the intended outcomes cannot be achieved by just identifying microaneurysms. Thus, KNN, SVMs, and ensemble-based techniques were used to offer more means of differentiating aneurysms and rating DR.

Retinal pictures are the main source of information used by computer-aided diagnosis to identify and classify diseases [5]. Nonetheless, the screening services for diabetic retinopathy—whether in developed or developing nations—are uneven and confront persistent obstacles, like the absence of precise recommendations regarding the best screening techniques (e.g., fundus photography versus clinical examination) and the growing cost of setting up and maintaining extensive DR screening programs. As a result, DR is becoming a more significant public health issue, particularly in many middle-class to low-income nations, where access to modern eye care treatments, such as laser and intravitreal therapies, and qualified eye care practitioners can be limited.

Moreover, it is essential to acknowledge that the manual procedure employed for conducting eye examinations is not only time-consuming but also a meticulous and labor-intensive endeavor, as highlighted in [6]. It is worth noting that diabetes and its associated complications impose a substantial economic burden on individuals and their families, as well as on national economies, mainly due to the mounting medical expenses and the resultant loss of wages. Consequently, the prospects of vision loss can be significantly reduced, possibly by as much as 90%, through the early identification of diabetes and its ensuing consequences. The chief structural modifications induced by DR manifest within the intricate layers of retinal tissues. The retina, an exceedingly thin layer comprised of nerve cells that possess light sensitivity, spans the posterior wall of the eye and is situated in close proximity to the optic nerve, as expounded upon in [7]. It is here that incoming light is absorbed by optic nerve cells, subsequently undergoing transformation into neural impulses. These impulses are then conveyed by the optic nerves to the brain, and our capacity to perceive and discern visual stimuli hinges upon the intricate processing of these signals by the brain. It is imperative to recognize that impairment of vision can arise from retinal damage triggered by a multitude of medical ailments.

A. RESEARCH OBJECTIVE

DR is an extremely severe medical condition, rendering its unfortunate victims permanently limited vision. The escalating prevalence of diabetes, driven by factors such as increased life expectancies and extravagant lifestyles, is a matter of growing concern. To ensure both treatment efficacy and cost-efficiency, individuals afflicted with diabetes and DR must receive timely and appropriate care. Nonetheless, initiating treatment is an intricate challenge due to the absence of symptoms until the disease has advanced to a later stage. Hence, our principal objective revolves around the early prediction of DR through the utilization of diverse methodologies, including cutting-edge deep learning and machine learning models. These models take as input a retinal image and produce an accurate assessment of the disease's stage. It is worth noting that the application of deep learning in the medical realm is accompanied by inherent complexities. Hence, there is a pressing need to ensure that these models and their outputs are presented in a comprehensible manner that is accessible to medical professionals, thus rendering artificial intelligence more interpretable in the domain of medicine. Furthermore, we employ an elucidative visualization technique known as gradient-weighted class activation mapping (Grad-CAM) to generate heat maps, ensuring that the rationale behind the model's decision-making can be intuitively communicated, whether directly or through the model's output. We have thoughtfully selected a range of models and subjected them to rigorous comparison, assessing their performance in terms of accuracy, precision, and recall scores. The successful early prediction of DR by a functioning model paves the way for the prompt referral of patients to ophthalmologists for further evaluation and

tailored care. The implementation of this kind of algorithm could greatly reduce the incidence of DR-related vision loss worldwide.

B. RESEARCH PROBLEM

In this study, we employ various methods in transfer learning, machine learning, and deep learning, in conjunction with the suggested model, and conduct a comparative evaluation. The dataset has certain constraints, notably an uneven distribution of classes, which could potentially affect the outcomes for both established and the proposed models. As a solution, we implemented a binary classification for the proposed model, which yielded outstanding results. Additionally, the scarcity of datasets specifically designed for research required our utilization of the Kaggle competition dataset.

C. CONTRIBUTION AND IMPACT

The contributions of this study can be summarized in the following manner:

- We introduce a novel customized CNN model that achieves an impressive 95.27% classification accuracy for both DR and No_DR. Advancements in deep learning models have significantly impacted various domains, particularly in the realm of medical image processing, notably enhancing tasks like DR diagnosis.
- Our study validates the efficacy of our dual approach, which includes the thorough comparison of diverse transfer learning models, encompassing both machine learning and deep learning models, in conjunction with our newly proposed model. This comprehensive methodology enables the identification of the optimal model for DR diagnosis, supported by a nuanced analysis using various metrics to evaluate the model's performance across multiple dimensions.
- The study leveraged a range of image pre-processing techniques, including Random Rotation, Horizontal and Vertical Flip, Resample, Constant Filling, and Zoom, to enhance the quality of retinal fundus images by eliminating artifacts and noise.
- Furthermore, an augmentation strategy was implemented to expand the image dataset, resulting in a substantial increase to 28,059 images in the augmented dataset.
- In addition, various optimizers such as Adam, Nadam, RmsProps, and stochastic gradient descent (SGD) were employed to determine the most suitable optimizer for the model, thus optimizing its performance.
- The utilization of the XAI technique Grad-CAM provided valuable visual insights into the predictions generated by the CNN model, facilitating the identification and highlighting of specific areas within an input image that significantly influence the ultimate classification or decision made by the network.
- This study encompassed the application of both single-label and multi-label classification techniques, utilizing established models such as SVM, RNN, K-nearest neighbors (KNN), DenseNet-121, Xception, ResNet-50, VGG-16, VGG-19, Inception, and our innovative model. It is worth mentioning that the proposed model has shown remarkable accuracy in multi-label classification scenarios. Additionally, it performed exceptionally well in single-label classification tasks with minimal ambiguity. To deepen our understanding of model behavior, we conducted experiments involving both pre-trained and randomly initialized weights in both of these methodologies.
- To further examine the capabilities of our proposed CNN model we have used two distinct datasets, Messidor2 and IDRiD, achieving impressive accuracies of 93.86% and 91.18% respectively. These results underscore the efficiency and effectiveness of the proposed model in diabetic retinopathy detection.

The remainder of the paper is structured as follows. Section II discusses some earlier studies in this field. Section III covers the proposed methodology, models used for experiments and provides a detailed examination of the datasets of the proposed system. Section IV provides the results of various experiments. Finally, Section V includes the conclusion and future development.

II. RELATED WORKS

DR is a microvascular consequence of diabetes that, if left untreated, can result in blindness. To avoid vision loss, early detection and treatment of DR are critical. The gold standard for DR screening is fundus photography, however, it is time-consuming and requires skilled individuals to interpret the results. There has been a surge of interest in the development of automated DR diagnosis solutions in recent years. These techniques examine fundus images and detect symptoms of DR using artificial intelligence (AI). Some AI-powered DR diagnosis systems have demonstrated impressive accuracy. Most AI-based DR diagnosis algorithms, on the other hand, have been built utilizing vast datasets of photos from specialized clinics or studies. These datasets may not accurately represent the population and the precision of AI-driven systems that detect retinopathy (DR), in real-world scenarios may vary. Multiple studies have evaluated the accuracy of AI-based DR diagnosis systems in real-world environments revealing that their performance can differ. For example, the study [8] found that an AI-based DR diagnosis system can achieve an accuracy of 90.6% in a real-world setting. However, in [9], research found that the accuracy of an AI-based DR diagnosis system dropped to 85%. Various factors can influence the accuracy of these systems, such as image quality, concurrent eye diseases, and the stage of DR itself. Generally speaking, AI-driven DR diagnosis systems tend to excel at identifying cases, then the early stages of DR.

Naive Bayes (NB) is a straightforward machine-learning technique for classification tasks. It operates by estimating the chance of a data point belonging to each class based on the class's prior probabilities and the likelihood of the data point assigned to each class. With an accuracy of roughly

85%, NB has been employed for DR diagnosis. It is a simple algorithm to implement that is also computationally efficient. However, NB is not very sensitive to changes in the data and can be overfitted.

The study examined a dataset of 5,000 fundus pictures, 2,500 of which were healthy and 2,500 of which had DR. In the test set, the NB classifier achieved an accuracy of 75.13%. SVM is a well-known model that may be used for classification as well as regression. SVM finds a hyperplane in the data that divides the data points into two groups. SVM has been used to detect DR with a reported accuracy of 90%. SVM is more resistant to data changes than NB and thus less prone to overfitting. However, training SVM can be computationally costly, particularly for large datasets. For example, in [10] suggested an SVM-based model for the identification of DR. A separate test set yielded an accuracy of 94.5% for the model, which was trained on a dataset of 10,000 fundus images.

To create a prediction that is more accurate, Random Forest (RF) aggregates the predictions made by several decision trees. Compared to individual decision trees (DTs), RF is more resistant to overfitting and can be utilized to uncover intricate correlations in the data. It has been stated that RF can detect DR with 91% accuracy. RF is computationally efficient and reasonably easy to implement. The study [11] demonstrates that RF is useful for DR diagnosis. 89.5% accuracy was attained by the model on a dataset including 200 rental photos.

An AI-based model called the bi-directional long short-term memory (BiLSTM) model can be used to identify and categorize various stages of dementia. The researchers employed a pre-processing technique called multiscale retinex with chromaticity preservation (MSRCP) to enhance the fundus pictures' contrast and quality prior to applying the LSTM model. This was done to increase the accuracy of the results and make the images easier for the LSTM model to interpret. The study [12] used the LSTM network to diagnose diseases with a 92% accuracy rate. In a similar vein, the study [13] presented a cutting-edge LSTM model to illustrate the effective use of KOA severity categorization and early detection based on the BiLSTM network. The algorithm achieved an accuracy of 78.57% in cross-validation and 84.09% in testing.

CNN is a type of neural network that works well for tasks involving image recognition. CNN has been utilized with over 93% accuracy for DR diagnosis. CNNs can be computationally expensive and difficult, though. CNN might often be challenging to understand. The study [14] used a radial basis function neural network to detect DR with an accuracy of 89.4%. Using a retrained AlexNet CNN, the study [15] obtained 92.5% accuracy for DR diagnosis in another investigation.

In the study [16], they employed a CNN model known as the DenseNet-169 classifier, which achieved an impressive accuracy rate of 90%. The authors utilized two datasets, namely the diabetic retinopathy detection 2015 and the APTOS 2019 Blindness Detection dataset obtained from Kaggle. These datasets contained images characterized by black backgrounds, black corners, and varying sizes, and all of these challenges were effectively addressed during the data pre-processing phase. Furthermore, to address the issue of imbalanced data, the authors implemented data augmentation techniques. Subsequently, using the prepared data, their proposed model showed an accuracy of 95% along with a validation accuracy of 90%. It should be noted that the reported accuracy surpasses that of other models used for comparison, including SVM which has an accuracy of 85.6%, regression that has an accuracy of 78%, Decision Tree (DT) with an accuracy of 85.1%, and KNN with an accuracy of 55.17%.

In the work described in reference [17], the researchers applied CNNs, Inception, VGG16, and ResNet to the task of classifying the different stages of DR. The dataset utilized for this study included 35,000 eye images with five distinct stages of DR, obtained from Kaggle, as well as the Messidor dataset, comprising 1,200 images representing four stages of DR progression. Notably, the VGG model exhibited the highest accuracy, achieving a 67% accuracy rate and an AUC (Area Under the Curve) of 0.67, establishing it as the most effective model. The researchers also carried out further optimization of the neural networks by incorporating different features such as attention maps and experimenting with numerous image pre-processing techniques. Likewise, in the study documented in [18], a machine learning approach was employed to anticipate whether an image contains any signs of DR, using data sourced from the UCI dataset. In this investigation, the sigmoid activation function was chosen for its soft-switching capabilities and straightforward derivative. The study reported promising results in the field of DR diagnosis, indicating the potential effectiveness of the machine learning approach.

Similarly, in the study [19], a twofold contribution was made. Firstly, they introduced a specialized neural network architecture designed for the task of DR image recognition, which demonstrated outstanding performance when compared to traditional feature extraction-based techniques. Additionally, their proposed algorithm benefited from the implementation of data augmentation, enhancing the algorithm's overall efficiency.

In the research conducted in [20], they developed a deep CNN designed for the early detection of DR by accurately identifying microaneurysms and appropriately labeling images of the retinal fundus, subsequently classifying them into five distinct categories. To optimize their implementation, the authors utilized a 4×4 Kernel-based CNN network in conjunction with an augmentation method. Their dataset comprised 88,702 images from the Kaggle EyePACS dataset, ensuring an equal distribution of images across all disease stages. Particularly, their model excelled in the low-high DR classification, achieving a sensitivity of

98% and a specificity of 94% for detection in the very early-stage. Additionally, they attained a quadratic weighted kappa score of 0.851 on the Kaggle dataset's severity scoring test set and an AUC (Area Under the Curve) score of 0.844. These impressive results validated the utility of their proposed model for clinical applications.

The research presented in [21] delves into the examination of two categories of DR: non-proliferative diabetic retinopathy (NPDR) and proliferative diabetic retinopathy (PDR). The study utilized image datasets sourced from diabetic retinopathy databases to train neural networks. Based on the training datasets, the goal was to classify individuals into four categories: (i) absence of diabetic retinopathy, (ii) mild nonproliferative diabetic retinopathy, (iii) severe nonproliferative diabetic retinopathy, and (iv) proliferative diabetic retinopathy. The proposed approach harnessed a CNN model. Image processing, employing SVMs (SVM), yielded an 88% recognition rate, while neural networks (NN) utilizing a local texture classifier based on multilayer perception achieved a recognition rate of 91%. The deep CNN, on the other hand, exhibited an impressive recognition rate of 93%.

The author [22] highlighted diagnosing diabetic retinopathy (DR) from fundus images is a specialty of deep learning. The goal of this work was to develop a highly precise and effective system for DR stage classification. A unique CNN model named RetNet-10 was created by pre-processing images, combining three datasets, and using data augmentation. RetNet-10 outperformed six state-of-the-art models and achieved an astounding 98.65% testing accuracy in the DR stage classification. RetNet-10 also shows resilience in a variety of k-fold validation tests. This work opens the door to better patient care by showcasing the potential of deep learning for precise and effective DR diagnosis.

Furthermore, in another research in [23] describe Effective therapy for DR, the primary cause of blindness in people with diabetes, depends on early identification, which requires time-consuming evaluations by ophthalmologists. For precise DR severity classification, this paper suggests a unique automated system called DR-CCTNet that makes use of a modified Compact Convolutional Transformer (CCT) model. DR-CCTNet achieved an amazing 90.17% test accuracy by combining five datasets into a large collection of 53,185 photos, applying image pre-processing and augmentation techniques, and efficiency-optimizing the CCT model. By far outperforming well-known transfer learning models such as VGG19 and ResNet50, this illustrates the robustness and promise of DR-CCTNet in transforming DR diagnosis, relieving load on ophthalmologists, and ultimately enhancing patient care.

III. METHODOLOGY

In this experimental study, we used transfer learning methods to enhance the performance and accuracy of six pre-trained models from the Keras library. The models selected were InceptionV3, VGG16, VGG19, ResNet50, Xception, and DenseNet-121. Fig 1 provides an overview of our proposed work with Explainable AI (XAI). We tested a collection of images from the Kaggle Aptos dataset on each of these models. The images, which represented five classes of diabetic retinopathy (DR) — 0 (No DR), 1 (Mild DR), 2 (Moderate DR), 3 (Severe DR), and 4 (Proliferative DR) — were processed by the pre-trained models to classify them into their respective categories. Following the experimentation, we performed a comprehensive comparison of the models to determine which one exhibited the highest levels of accuracy, precision, and recall based on the results obtained. We used Tensorflow ML framework and the version is 2.14.0. Additionally, we incorporated Scikit-learn and Keras into our methodology for training and generating valuable results.

Experimental Setup Specifications: The study employed an AMD Ryzen 7 5800H processor with 8 cores, 16 threads, and a 3.2 GHz clock speed. The system is equipped with 16 GB of RAM and a 512GB m.2 Gen3 SSD for all experiments. Additionally, it features an Nvidia GeForce RTX 3060 GPU with 6 GB of video RAM (VRAM). The setup utilizes the Nvidia Cuda Toolkit version 12.2, TensorFlow 2.14.0, and Jupyter Notebook version 7.0.4 as the integrated development environment (IDE).

Data Processing: This study used the 'APTOS 2019 Blindness Detection' dataset, sourced from Kaggle, which encompasses 3,662 high-resolution color fundus retinal images with associated tags. The dataset is publicly available in the link https://www.kaggle.com/competitions/aptos2019-blindness-detection/data. Since these images were collected with multiple cameras over an extended period of time from different clinics, there is a great deal of variation and inconsistency in the aspect ratio, image quality, and other aspects of the images, as highlighted in a study in [24].

Within this dataset, each of the five classes corresponds to a distinct stage of the disease, categorized on a severity scale ranging from 0 to 4, as elucidated in [16]. Specifically, among the 3,662 ocular images, there are 1,805 images representing the absence of dry eye disease (DR), 999 images with moderate manifestations, 370 images indicating mild severity, 295 images reflecting proliferative conditions, and 193 images depicting severe stages of the disease. The description of different stages of diabetic retinopathy is given below.

- No diabetic retinopathy- This condition indicates the absence of any form of diabetic retinopathy, signifying a perfectly healthy eye without any damage to the retina.
- Mild diabetic retinopathy- At this early stage of diabetic retinopathy, there is a noticeable expansion of the blood vessels within the eye's retina, a phenomenon referred to as microaneurysms. In addition, a small amount of fluid can seep into the retina, resulting in enlargement of the macula, a critical region of the retina located at its center.
- Moderate diabetic retinopathy - In the second stage of diabetic retinopathy, inflammation affects the blood

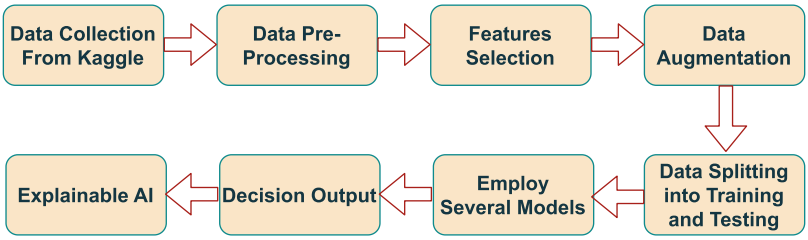


FIGURE 1. Top-level overview of our proposed work with XAI.

TABLE 1. Data distribution across the dataset.

Class Labels	Number of Images
No DR	1805
Mild	370
Moderate	999
Proliferate DR	295
Severe	193

vessels, impeding the normal blood supply to the retina. This blockage leads to the entrapment of essential nutrients and other supplies, causing the macula to become obscured by a buildup of blood and other fluids.

- Severe diabetic retinopathy- This stage is characterized by more intense vascular occlusion, affecting a larger portion of the retinal blood vessels. This extensive occlusion results in reduced blood flow to various parts of the retina, signaling the body to initiate the development of new blood vessels.
- Proliferative diabetic retinopathy- Representing the most severe stage of diabetic retinopathy, this condition involves excessive growth of blood vessels within the retina. Weaker blood vessels at this stage are prone to fluid leakage, leading to various vision-related complications, including blurriness, narrowing of the field of vision, and, in severe cases, potential blindness.

Table 1 represents the data distribution across the dataset, the dataset contains labels along with the corresponding sample counts. The distribution of samples in the dataset is uneven; it should be emphasized that the data quality is of exceptional standard. Consequently, our strategy is to present an advanced methodology without the need for additional data augmentation.

Image Transformation: Edge detection, semantic segmentation, and grayscale image colorization are among the numerous applications based on image transformation tasks, as discussed in [25]. These tasks involve processing images where the input and output are visual data. Pre-processing is necessary to prepare image data for model input, as highlighted in the work in [26]. The pre-processing of images not only streamlines the time required for model training but also expedites the model's inference process. Resizing input images, for example, may not impact the model's performance, however it can significantly reduce the time needed for training, particularly when dealing with excessively large images. To select the most appropriate pre-processing techniques for enhancing model performance, a comprehensive understanding of the problem, the collected data, and the production environment is indispensable. In our study, we used a range of image manipulation methods to achieve the goal.

i) Image flipping: Image flipping involves the rotation of a picture along either a horizontal or vertical axis. This technique introduces randomness in the flipping process, allowing the model to learn to recognize and correctly label flipped images. As a result, when the model encounters a horizontally or vertically flipped image, even if it pertains to a challenging case such as a '3 - Severe DR' classification, it can identify and label the image accurately, ensuring there are no irregularities or inconsistencies.
   - Horizontal flipping: Horizontal flipping is reversing an image's full row and column pixels horizontally (left to right) where the flipping happens on the vertical axis.
   - Vertical flipping: Vertical flipping involves reversing the order of pixels in each row and column, creating an upside-down version of the original image as the flipping action occurs along the horizontal axis.
   - Random Rotation: The random rotation augmentation method introduces variability by randomly rotating images in a clockwise direction. This rotation can range from 0 to 360 degrees, ensuring a diverse set of image orientations for training and evaluation.

ii) Resample=Image.LANCZOS: Employing the Lanczos interpolation method yields a more refined and sharper image as opposed to a blurry one. When it comes to preserving intricate details, reducing aliasing effects, and minimizing artifacts, Lanczos interpolation is considered the most optimal choice. This method involves filtering the input samples from the original image through the Lanczos kernel, effectively reconstructing the image. However, it is important to note that along the sharper edges of the resulting image, there may be

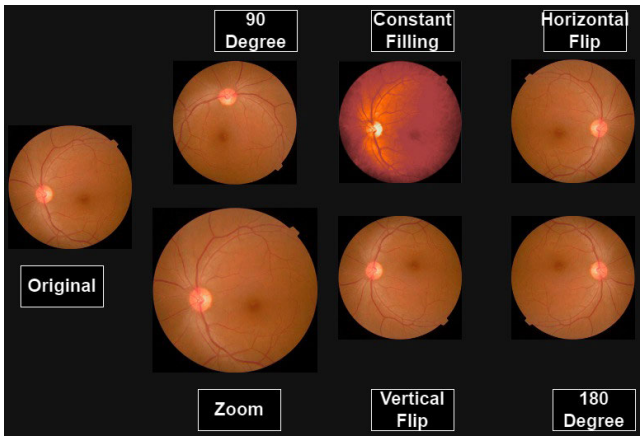


FIGURE 2. Dataset after augmentation process.

a presence of dark or bright halos, contributing to the perceived image sharpness.

iii) Image Zoom: Image zooming entails enlarging an image to enhance the visibility and clarity of its features. Zooming allows for the elimination of extraneous areas, enabling the model to concentrate on the relevant regions of interest. When it comes to precisely categorizing the severity of eye damage, focusing on specific regions within the image becomes critical. Every pixel's value plays a role in model training during prediction; thus, minimizing the influence of pixels beyond the image border leads to more accurate classifications.

iv) Constant Filling: To ensure a clean and uncluttered image classification process, it is imperative to eliminate the unnecessary black regions surrounding the retinal images. This adjustment is crucial because dark areas in the images could potentially impact the model's predictions, and it is essential to exclude them from consideration.

Dataset Augmentation: In the Aptos datasets, we have addressed the issue of imbalanced labels by implementing data augmentation. The distribution of images across different labels is uneven, with 'No DR' having the highest number of images and 'Moderate' being the second highest. Conversely, the 'Severe' label has a notably low count of 193 images. To mitigate this imbalance, we applied various data augmentation techniques, encompassing six types, including Random Rotation (0 degrees to 360 degrees), Horizontal Flip, Vertical Flip, Resample, Constant filling, and Zoom. Fig. 2 shows all types of augmentation processes of the dataset. Following augmentation, we also increased the image for No DR 3 times, Mild 15 times, Moderate 6 times, Severe 30 times and PDR 16 times so that after augmentation, each of the label images will be almost the same length, and finally, the total number of images in our dataset increased to 28,059. Table 2 provides an overall idea of before and after augmentation of our dataset.

We have made sure that the processed images maintain their visual and semantic significance for any other interpretation by humans. Pre-processing techniques have been selectively applied to certain sections of the image as needed, while leaving other areas unaffected. To achieve the best possible image quality, we have used several iterations to fine-tune the pre-processing stages. Additionally, since testing on a variety of photos guarantees that the pre-processing stages are robust across numerous scenarios, we have routinely confirmed the impact of pre-processing on a portion of images and evaluated the overall impact on the tasks.

TABLE 2. Before and after data augmentation.

Class Labels	Before Augmentation	After Augmentation	Increased By
No DR	1805	5414	3
Mild	370	5550	15
Moderate	999	5994	6
Proliferate DR	295	5310	30
Severe	193	5790	16
Total	**3662**	**28059**	

Data Split: The training set serves as the foundation for estimating parameters, comparing several models, and executing the necessary activities to achieve the objective of the final model. This process involves the development of models and features [27]. After the completion of these activities, the test set is utilized to assess the performance of the model. The objective is to apply the model in practical applications, where it is anticipated to effectively align with available data when both inputs and outputs are known, and reliably forecast the outcomes of unknown data [27].

In our study, a total of 28059 photographs were utilized for training purposes, whereas 1928 images were allocated for validation. By increasing the quantity of photographs used for training, the model will acquire the ability to accurately identify and classify images, therefore enabling it to provide noteworthy results upon completion of the testing phase. To enhance the performance of the proposed model, we used a pre-trained model, which facilitated the differentiation between infected and healthy eyes during the image processing phase. This approach led to outstanding training and validation accuracy.

A. TRANSFER LEARNING

1) DenseNet-121

The primary objective of the DenseNet design, as proposed in [28], is to improve the depth of deep learning networks. By implementing shorter connections between layers, the network's performance is enhanced, and the computational cost associated with training is reduced. DenseNet consists of numerous interconnected layers, with each layer fully linked to all subsequent levels, allowing for optimal data transmission across different network levels. Inputs from all preceding layers are distributed to each layer, and feature mappings are propagated to all following levels to preserve a feed-forward architecture. DenseNet-121, as described in [29], is structured into Dense Blocks, where the feature

size remains constant within each block, while the number of filters varies across the blocks.

2) XCEPTION

As explained in [30], Xception is comprised of a linear series of separable depth-wise convolution layers linked by residual connections. The main goal of depthwise separable convolution is to minimize computational demands and memory consumption. The primary objective of using depth-wise separable convolution is to reduce both memory usage and computational requirements. These layers are organized into 14 modules, with 36 convolutional layers in total, all of which incorporate linear residual connections, except for the initial and final layers. Within the Xception architecture, separable convolution separates the learning of channel-specific and spatial features. The Xception model, as outlined in the work in [31], can be divided into three main sections: the entry flow, middle flow, and exit flow. Image data is first processed through the entry flow, followed by eight iterations of the middle flow, and concludes with the exit flow. Furthermore, the inclusion of residual links, as proposed in [32], introduces shortcuts in the sequential network, effectively addressing the problem of vanishing gradient and representation bottlenecks.

3) InceptionV3

The Inception V3 CNN model, described in [33], is a widely recognized model for image recognition and object detection. Its credibility is demonstrated by achieving an accuracy rate of over 78.1% on the ImageNet dataset. This CNN model is characterized by its depth of 27 layers, and it includes an inception layer that combines 1×1 convolutional layers, 3×3 convolutional layers, and 5×5 convolutional layers. The output filter banks of each of these layers merge into distinct output vectors, which then serve as inputs for the subsequent phase. Additionally, a 1×1 convolutional layer is introduced to reduce dimensionality, and the max-pooling layer is retained as an alternative to the inception layer. The Inception layer enables the internal layers to select the most suitable filter size for their specific tasks.

4) VGG-16

VGG-16, developed in 2014 with the aim of winning the ILSVRC (Imagenet) competition, achieving an accuracy of 92.7% using the ImageNet dataset, as reported in [34], which contains a vast dataset of 14 million images. What separates VGG-16 from other models is its focus on utilizing 3×3 convolutional layers with a stride of 1, consistent padding, and 2×2 max-pooling layers with a stride of 2 throughout its entire architecture. This design choice minimizes the need for a multitude of hyperparameters. With a total of 16 layers incorporating weights, VGG-16 is a substantial network with approximately 138 million parameters. The input images undergo a sequence of convolutional layers, followed by spatial pooling, and culminate in max-pooling. The final segment includes two fully connected layers preceding a softmax layer for output. All the convolutional layers employ rectification non-linearity (ReLU) without local response normalization (LRN), resulting in efficient memory usage and faster computation times.

5) VGG-19

VGG-19 consists of multiple layers, including 16 convolutional layers, 5 max-pooling layers, 3 fully connected layers, and 1 softmax layer. It begins with the input of a fixed-size image, followed by pre-processing. Spatial padding is employed to maintain spatial resolution. Next, the model performs max-pooling using 2×2-pixel windows with a stride of 2. Subsequently, the Rectified Linear Unit (ReLU) is applied to introduce nonlinearity, enhancing VGG-19's image classification capabilities and improving computational efficiency. The model outperforms previous ones that used tanh and sigmoid functions. Although VGG-19 demands more memory usage, it is considered superior in terms of performance.

6) ResNet-50

ResNet-50, a CNN, is composed of 50 layers. It includes 48 convolutional layers with 64 different kernels, as explained in [18]. Additionally, there is one max-pooling layer with a stride of 2, and these layers are replicated three times to yield a total of nine layers. The subsequent layer introduces different kernels and is repeated four times, resulting in a total of 12 layers. Various kernel variations are employed in the following layers, which are repeated multiple times, culminating in 49 layers. The architecture concludes with an average pooling layer and a densely connected layer comprising 1,000 nodes, followed by a softmax function, which serves as the final layer. The model can also be loaded with a pre-trained version, trained on images from the ImageNet database, providing the network with a wealth of knowledge in feature representation for a wide range of images.

B. MACHINE LEARNING AND DEEP LEARNING MODELS

1) RECURRENT NEURAL NETWORK (RNN)

RNN has been designed specifically to process sequential data by effectively preserving information from previous inputs. It finds use in several tasks such as time series analysis, natural language processing (NLP), and some image processing applications, enabling the retention of information across consecutive time steps. RNNs have demonstrated use in medical domains, namely in the analysis of patient history, disease development, and therapy response, as exemplified by their application in diabetic retinopathy. The researchers employ sequential patient data, including glucose levels and clinical records, to ascertain the likelihood of developing a condition. RNNs that incorporate attention processes are designed to prioritize visual features and integrate information from many sources, therefore providing a comprehensive perspective on the health status of a patient.

2) K-NEAREST NEIGHBORS

The KNN algorithm is a popular machine learning technique employed in the examination and evaluation of diabetic retinopathy. It is a non-parametric method that does not rely on underlying data assumptions. Instead, it evaluates the similarity of the data points, which makes it suitable for various tasks in the detection of diabetic retinopathy. KNN can compute relevant features from retinal images, such as texture characteristics, color histograms, or image statistics. These features encapsulate critical image attributes and can aid in disease detection. For straightforward image classification tasks related to diabetic retinopathy, especially when dealing with limited datasets and computational resources, KNN may be a useful choice, as it assesses the similarity of retinal images to make classification predictions.

3) SUPPORT VECTOR MACHINE

The SVM is a powerful machine learning algorithm widely used for classification and regression tasks. It excels in handling high-dimensional data and can address non-linear separations through kernel functions. In image classification, SVM is valuable for extracting features from retinal images, identifying characteristics such as vessel patterns and texture characteristics, and aiding in the diagnosis and categorization of diabetic retinopathy. SVM can also complement other machine learning or deep learning models and pre-processing techniques, contributing to accurate diagnosis and patient care in the detection of diabetic retinopathy.

4) CONVOLUTIONAL NEURAL NETWORK (CNN)

CNN stands for Convolutional Neural Network, which is a specialized deep learning model primarily designed for structured grid-like data, particularly images. In the context of diabetic retinopathy detection, CNNs demonstrate remarkable proficiency in the process of Feature Extraction. They possess the capability to automatically identify anomalies such as lesions and microaneurysms in retinal images. Additionally, CNNs can effectively categorize retinal images into different severity stages of diabetic retinopathy, thus facilitating early diagnosis and risk assessment. Notably, CNNs display the valuable trait of translation invariance, allowing them to identify objects within images irrespective of their position, a crucial attribute for tasks involving image analysis. The utilization of pre-trained CNN models, fine-tuned specifically for diabetic retinopathy, mitigates the necessity for extensive data acquisition and training, rendering them indispensable tools in the realm of medical image analysis.

C. PROPOSED MODEL

Our proposed model features a 5-layer Conv2D architecture, gradually increasing channel sizes from 8 to 128, utilizing 3×3 to 5×5 kernels with a stride of 1. ReLU activation is applied, enhancing gradient propagation. Two max-pooling layers reduce dimensions, from 224 to 112 and then to 56. A dropout layer with a 50% rate addresses overfitting after the second max-pooling. Batch normalization follows, ensuring standardized input. The output is flattened for dense layers, with ReLU activation, promoting efficient gradient descent. The final dense layer, which serves as the classification layer, employs softmax activation for output normalization, crucial for converting weighted sums into meaningful probabilities.

1) CONVOLUTION LAYER

The most basic operation for applying a filter to an input to produce an activation is known as convolution, as explained in [35]. When the same filter is applied to a single input multiple times, it generates a feature map that reveals the positions and strength of a recognized feature in the input, such as an image. In the proposed model (Fig. 3), we utilized a 2D convolutional layer, often abbreviated as conv2D. The layer moves the kernel across the 2D input data, performing element-wise multiplications, and then consolidates all the results into a single output pixel. This process is repeated each time the kernel traverses a location, transforming the 2D feature matrix into a different 2D feature matrix.

In the proposed model, we have used five convolutional layers, each followed by a max-pooling layer. The first conv2D layer consists of 8 input channels and employs a 3×3 kernel with a stride of 1. Subsequently, we incrementally increased the layer size to reach a maximum of 128 channels using a 5×5 kernel, which results in a 5×5 filter matrix. The filter moves through the input volume with a one-unit step as the stride is set to 1. The activation function applied in these layers is ReLU, which offers several advantages over traditional units, including effective gradient propagation and faster computational performance compared to sigmoid units. After the initial Conv2D layer, the output is passed to a MaxPooling2D layer, and the output from the max-pooling block proceeds to another convolutional block, where the number of channels is further increased to 128.

2) MAX-POOLING LAYER

The max-pooling layer serves to reduce the dimensionality of images by decreasing the number of pixels in the output of the preceding convolutional layer. This reduction is necessary because a large number of pixels would result in a higher parameter count, potentially leading to an excessive amount of data. The max-pooling operation involves selecting the most significant element within a region of the filter's feature map. Consequently, the output of the max-pooling layer is a feature map that retains the most critical features of the preceding feature map.

In the proposed model, we incorporate two max-pooling layers. The first max-pooling layer reduces the height and width of the image it receives from the initial conv2D layer, taking it from 224 to 112. The output is then forwarded to the second conv2D layer, which in turn feeds it into the subsequent max-pooling layer, further reducing the height and width to 56. Importantly, the number of filters in

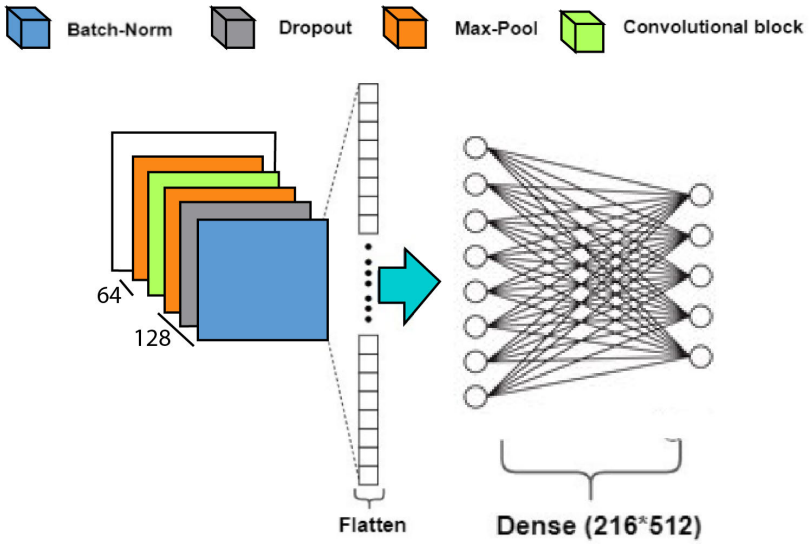


FIGURE 3. Proposed model architecture.

the max-pooling layers remains unchanged throughout this process.

3) DROPOUT LAYER

Large neural networks, as discussed in [36], are susceptible to overfitting because they involve stacking many layers to learn complex functions. Overfitting, described in [37], occurs when the model attempts to learn an excessive number of features from the training dataset, leading to poor performance and an inability to generalize the training data's patterns. To tackle this problem, the dropout technique, as explained in [38], is employed. Dropout serves as a regularization method that emulates the simultaneous training of numerous neural networks with different architectures. By temporarily removing a percentage of units, as outlined in [36], along with all their incoming and outgoing connections, overfitting is significantly reduced, offering substantial improvements over other regularization methods. In proposed model, the output from the second max-pooling layer is passed to a dropout layer with a dropout rate of 0.5, meaning that during each epoch, 50% of the neurons in this layer are randomly deactivated.

4) BATCH-NORMALIZATION LAYER

When input data is fed into a deep learning system, as discussed in [39], it is common practice to adjust the values to a standardized scale. The purpose of this normalization is to ensure that the proposed model can generalize effectively. The addition of extra layers to deep neural networks contributes to their speed and stability through batch normalization, as elucidated in [40]. Batch normalization acts as a network layer that promotes greater independence for each layer, normalizing the output of the preceding layers. In the proposed model, the output of the dropout layer serves as the input to the batch normalization block, where activation scales are applied to normalize it. Without going into technical details, the computation of batch normalization would occur as described in [36]:

$$z = g(w, x) + b; \quad a = f(z) \tag{1}$$

Now, when incorporating batch normalization into the equation, where '$g()$' represents the linear transformation of the neuron, 'w' denotes its weight, 'b' stands for its bias, and '$f()$' represents its activation function.

$$z = g(w, x); z^N = \left(\frac{z - m_z}{s_z}\right) \cdot \gamma + \beta; \quad a = f\left(z^N\right) \tag{2}$$

In this context, z^N corresponds to the output from the Batch Normalization layer, m_z represents the output of the neuron, s_z signifies the standard deviation of the neuron's output, and γ and β are the trainable parameters of the Batch Normalization.

5) FLATTEN LAYER

Flattening, as described in [41], is the procedure of transforming data into a one-dimensional array, allowing its use in the following layer. The flattened layer receives the output from the batch normalization layer and converts it into a one-dimensional array, which can then be input into the dense layers.

6) DENSE LAYER

The dense layer as discussed in [42] is a deep neural network layer. In this layer, each neuron receives input from all neurons in the preceding layer. Within the proposed model, the first two dense layers are succeeded by a dropout layer. The activation function employed here is ReLU, which effectively sets all negative activations to zero, greatly expediting the gradient descent process. The last dense layer, which also serves as the classification layer, employs the softmax activation function. Softmax explained in [43], is a

mathematical function that transforms a numerical vector into a probability vector, where the probability of each value is directly proportional to its relative magnitude within the vector. A primary advantage of utilizing Softmax is its ability to normalize the output, converting weighted sum values into probabilities that sum up to one.

D. MODEL TRAINING

In the experimentation involving all models, including our proposed CNN model, two distinct image sizes were employed—32 × 32 and 16 × 16. The training of each model incorporated an early stopping mechanism to prevent overfitting or underfitting. On average, 50 epochs were executed for each model, encompassing our proposed model. The duration of each epoch amounted to approximately 20 seconds for the 32 × 32 image size, while for the 16 × 16 image size, the average duration per epoch reduced to 10 seconds.

E. MODEL COMPILATION

1) EVALUATION METRICS

Accuracy serves as a vital evaluation metric across various domains, especially in the realm of classification tasks, including machine learning, deep learning, and data analysis. Its significance is multifaceted. It provides a quantifiable gauge of a model's efficacy, offering a clear reflection of its competence in accurately classifying data points. Accuracy, as defined in [44], is a measure that evaluates a model's performance and overall effectiveness across all classes. It becomes particularly useful when all classes hold equal importance. It is calculated by determining the ratio of correct predictions to the total number of predictions. However, it is important to note that accurate measurement relies on having an equitable distribution of samples in each class.

$$\text{Accuracy} = \frac{\text{True}_{\text{pos}} + \text{True}_{\text{neg}}}{\text{True}_{\text{pos}} + \text{True}_{\text{neg}} + \text{False}_{\text{pos}} + \text{False}_{\text{neg}}} \tag{3}$$

Precision stands as a crucial assessment metric, predominantly applied in the realm of classification problems, with the aim of gauging the model's accuracy in making positive predictions. Precision, as defined in [44], serves as a gauge for the degree to which a model categorizes a sample as positive. When the model produces an extensive number of erroneous positive predictions or a limited number of accurate positive predictions, the denominator increases while accuracy decreases. In contrast, when the model achieves a significant proportion of correct positive predictions and fewer incorrect positive predictions, precision is high, emphasizing the maximization of true positives. This elevated precision indicates that false positives should be avoided due to their potential consequences. Precision proves to be particularly crucial in situations where false positives entail significant costs or ramifications. A notable example lies in the field of medical diagnosis, where a high-precision model is of paramount importance. In the medical context, misdiagnosing a healthy patient as having a disease can lead to unwarranted stress, expenses, and unnecessary medical treatments.

$$\text{Precision} = \frac{\text{True}_{\text{pos}}}{\text{True}_{\text{pos}} + \text{False}_{\text{pos}}} \tag{4}$$

The recall [44], also referred to as Sensitivity or True Positive Rate, serves as an evaluation metric primarily employed in the context of classification tasks. Its primary purpose is to assess a model's proficiency in correctly recognizing all pertinent instances of a specific class. In essence, recall quantifies the fraction of actual positive instances that the model accurately categorizes as positive. The higher the recall value, the more positive samples are successfully detected. It is worth noting that recall exclusively pertains to the correct classification of positive samples. Recall holds particular significance in scenarios where overlooking a positive instance can have substantial repercussions. For instance, in the realm of medical diagnostics, recall assumes critical importance. Failing to identify a disease when it is indeed present can lead to severe health-related consequences.

$$\text{Recall} = \frac{\text{True}_{\text{pos}}}{\text{True}_{\text{pos}} + \text{False}_{\text{neg}}} \tag{5}$$

2) OPTIMIZER

Adam, (Adaptive Moment Estimation), is a widely used optimization algorithm in machine learning, especially in the training of deep neural networks. It extends the SGD method and incorporates elements from both the RMSprop and momentum techniques. The Adam optimizer, as discussed in [45], is particularly well-suited for tasks involving large datasets and/or numerous parameters due to its computational efficiency, low memory requirements, and its ability to handle variations in gradient scales. One of Adam's key features is its adaptation of individual learning rates for each parameter using adaptive learning rate models. It leverages information from both the first and second gradient moments to dynamically adjust the learning rate for updating neural network weights. This adaptability is particularly beneficial in situations where Adam performs well, such as optimizing flat surface error minima.

Nesterov-accelerated adaptive moment estimation (Nadam) [46], a popular optimization algorithm for training deep neural networks, including CNNs, accelerates the process of transfer learning. This expedited adaptation minimizes loss, saving time and computational resources. It leverages knowledge from the pre-training dataset to adapt to new data, such as diabetic retinopathy, enhancing generalization and feature capture precision for accurate results in this medical application.

Root mean square propagation (RMSprop) [46], commonly used in deep neural network training including CNNs, optimizes learning rates for efficient transfer learning. By normalizing gradients and enabling adaptive learning rates, it expedites model adaptation to a new DR dataset while

retaining pre-training knowledge, enhancing generalization, and reducing manual hyperparameter tuning.

The term SGD [47] is a fundamental optimization technique in deep neural network training, known for its granular control over layer-specific learning rates. This level of fine-tuning is invaluable when adapting pre-trained models to new tasks. Additionally, SGD is highly effective for regularization, preventing overfitting through methods such as L2 regularization. Its adaptability extends to custom learning rate schedules, allowing adjustments during training, which is advantageous when fine-tuning fresh datasets with changing learning rate needs. In addition, SGD stands out for its computational efficiency, making it a suitable choice even in resource-constrained settings.

We have experimented with various optimizers to assess which one yields consistent and robust performance for our CNN model. Table 3 shows the experimental results of the precision of different optimizers.

TABLE 3. Experimental results of different optimizers regarding accuracy (%) of the proposed model.

Optimizer	Accuracy
Adam	**95.27**
Nadam	93.88
RmsProps	88.23
SGD	83.60

3) LOSS FUNCTION

A loss function, often referred to as a cost function or objective function, holds a central role in the training of both machine learning and deep learning models. Its primary function is to measure the extent of error or disparity between the model's predicted values and the true target values within the training data. Essentially, the loss function quantifies the model's performance by indicating how well or poorly it is performing. When the model's predictions closely match the actual data, the loss is minimized, and conversely, it increases when predictions deviate from the real values.

In proposed model, we employed two distinct loss functions: categorical cross-entropy for single-label tasks and binary cross-entropy for multi-label tasks. In binary cross-entropy, each predicted probability is compared to the actual binary class result, typically denoted as 0 or 1. The score is then determined by penalizing probabilities based on their deviation from the true values, reflecting how close or distant the predicted values are from the actual ones. For the calculation of binary cross-entropy in multi-label classification problems, the specific formula as described in [48] can be used.

$$\text{logloss} = -\frac{1}{N}\sum_{i}^{N}\sum_{j}^{M} y_{ij}\log\left(p_{ij}\right) \tag{6}$$

where,

- N: number of rows
- M: number of classes

Given the model's larger number of classes, we employ categorical cross-entropy for single-label classification. Categorical cross-entropy assesses the alignment between the predicted distribution (output layer activations) and the actual distribution, where the probability of the correct class is either 1 or 0 for the remaining classes. A lower score signifies improved model performance. The specific formula for calculating categorical cross-entropy for single-label classification is provided [49]:

$$L(y,\hat{y}) = -\sum_{j=0}^{M}\sum_{i=0}^{N}\left(y_{ij} * \log\left(\hat{y}_{ij}\right)\right) \tag{7}$$

IV. RESULTS

The most effective technique for diabetic retinopathy detection has been found using a two-fold technique that compares transfer learning models with the proposed model. Utilizing diverse metrics for model evaluation allowed for a more comprehensive assessment across various aspects.

For the first method, we used the models that were previously mentioned along with multi-label classification. To be more precise, we combined these models with a dense layer that had a *sigmoid* activation function, which allowed us to give each label a score between 0 and 1. However, this approach is less effective in early detection of DR. To be precise, the model can predict values greater than 0.5 for labels 0, 1, and 2, respectively, if the actual label for a sample is 2. Consequently, the model assigns different values to each label, and the label with the highest value is considered the model's prediction. Although this approach excels in multi-label classification with very high accuracy, its performance is less reliable when it is transitioned to single-label classification, resulting in a significant decrease in accuracy.

In contrast, employing the *softmax* activation function and *categorical crossentropy* for single-label classification has yielded promising outcomes in various models, including Xception and Inception. However, the proposed model stands out in terms of performance, surpassing the accuracy of Xception, which we initially identified as the most suitable model for this research. Specifically, Xception and Inception achieved accuracies of 82.18% and 80.91% respectively. On the contrary, the suggested model for single-label classification achieved a remarkable accuracy of 95.27%, surpassing the performance of the other transfer learning models examined in this study.

In both approaches, we utilized a combination of pre-trained and randomly initialized weights in order to enhance our understanding of the models' performance. The model was set using a range of parameters, including accuracy, precision, and recall. Additionally, the *Adam* optimizer was implemented to improve the efficiency of the model training procedure. We use an early stopping function

to reduce overfitting and underfitting. On average we use 50 epochs for each of the models.

A. RESULTS FOR MULTI-CLASS CLASSIFICATION

According to Table 4, for multi-class classification, the comparison table outlines the performance metrics of various deep learning models, including ResNet50, VGG16, Xception, DenseNet121, Inception, VGG19, and a proposed CNN model, with regard to trainable parameters, recall, accuracy, and precision, with results distinguished between random weight initialization and ImageNet weights initialization. ResNet50 emerges as a strong performer, particularly with ImageNet weights, showing a recall of 0.7655, accuracy of 0.7764, and precision of 0.7989, all markedly higher than those achieved with random weights. Xception also shows a notable improvement with ImageNet weights, achieving a recall of 0.7291, accuracy of 0.8218, and precision of 0.8833. Conversely, DenseNet121 struggles across all metrics, regardless of weight initialization, with the lowest values observed among all models. It is notable that the proposed CNN model, which surpasses all others, has the highest recall of 0.9585, accuracy of 0.9527, and precision of 0.9612, although with the largest number of trainable parameters at 103104901. These findings underscore the significance of weight initialization and model architecture in determining the efficacy of deep learning models, with the proposed CNN model emerging as the most promising option for classification tasks.

We evaluated various deep learning and machine learning models, including RNN, KNN, SVM, and CNN, to ensure a comprehensive comparison. Table 5 demonstrates classification accuracy, precision, and recall for machine & deep learning models. Through extensive training with different parameters and epochs, we observed that CNN models outperformed others, achieving an accuracy of 80.46%.

However, this result underscores the superior performance of the proposed model compared to other transfer learning and deep learning models. Comparatively, the majority of the models performed satisfactorily for multi-label classification; however, VGG-19 did not yield results that were sufficient for the detection of DR.

The proposed CNN model works best for all types of DR and No DR diagnosis, accurately classifying most of the samples, as shown in Fig. 4. On the other hand, among all transfer learning models, Xception performed well, but not as robust as the proposed CNN model. The majority of the samples with the DR labels 'severe' and 'proliferative' faced the problems. The unequal distribution of data is the cause of the uncertainty in identifying the sample labels.

Table 1 illustrates that the quantity of samples corresponding to severe and proliferative cases of DR is notably lower in comparison to other sample categories. Consequently, the model places more emphasis on samples labeled as 'Moderate' and 'No_DR' leading to relatively high accuracy, precision, and recall values for these classes. Therefore, we applied various image processing and augmentation

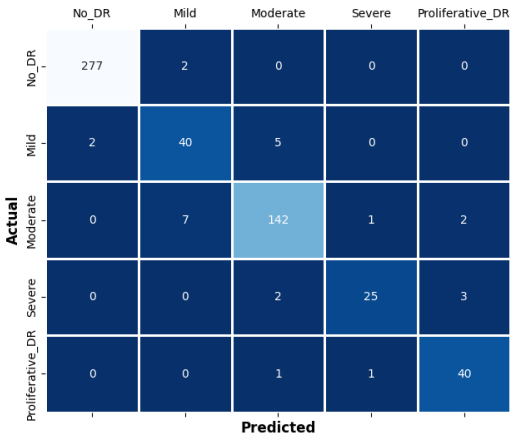


FIGURE 4. Confusion matrix of the proposed CNN model.

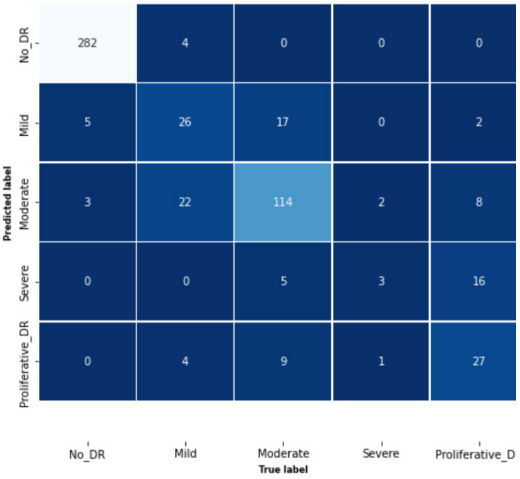


FIGURE 5. Confusion matrix of Xception.

techniques to make the dataset balanced; however, after augmentation still rest of the transfer learning models performance not close to our proposed CNN model. Fig. 5 displays the confusion matrix for the Xception model, revealing that a substantial portion of the 550 test samples is correctly predicted. The model tends to make more errors when handling samples with 'Proliferative DR' and 'Severe' labels.

The illustration in Fig. 6 vividly demonstrates the precise detection of multi-class DR, encompassing categories like Mild, Moderate, Severe, Proliferative_DR, and No_DR. This exemplary performance distinctly highlights the superiority of the proposed model when compared to pre-trained transfer learning, machine learning, and deep learning models.

B. RESULTS FOR BINARY CLASSIFICATION

To analyze the performance of the proposed model and other models used for comparison, we conducted binary classification. We grouped all cases (mild, moderate, severe, and proliferative_dr) as DR and designated the rest as No_DR.

TABLE 4. Comparison of transfer learning models trained for multi-class classification.

Model	Attributes	Trainable Parameters	Recall	Accuracy	Precision
ResNet50	random weights	23797509	0.5873	0.7345	0.873
	Imagenet weights	23797509	0.7655	0.7764	0.7989
VGG16	Random weights	14780997	0.4691	0.7491	0.9348
	Imagenet weights	14780997	0.4564	0.7455	0.9472
Xception	Random weights	21069869	0.62	0.7491	0.8525
	Imagenet weights	21069869	0.7291	0.8218	0.8833
DenseNet121	Random weights	7085701	0.4545	0.5873	0.6596
	Imagenet weights	7085701	0.0764	0.0764	0.0785
Inception	Random weights	22031269	0.4927	0.7382	0.9125
	Imagenet weights	22031269	0.7727	0.8091	0.8603
VGG19	Random weights	20090693	0.4527	0.7491	0.954
	Imagenet weights	20090693	0.0018	0.52	1
CNN	Proposed	103104901	0.9585	0.9527	0.9612

TABLE 5. Comparison of machine and deep learning models trained for multi-class classification.

Model	Recall	Accuracy	Precision
RNN	0.7851	0.7962	0.8233
KNN	0.6937	0.6211	0.8568
SVM	0.5602	0.5923	0.6315
CNN	0.8187	0.8046	0.8509

Table 6 presents the accuracy, precision, and recall metrics for binary label classification. The provided table offers insights into the performance of various transfer learning models trained for binary classification tasks, including ResNet50, VGG16, Xception, DenseNet121, Inception, VGG19, and a proposed CNN model. Evaluation metrics such as trainable parameters, recall, accuracy, and precision are reported, distinguishing between models initialized with random weights and those with ImageNet weights. ResNet50 demonstrates notable improvements with ImageNet weights, achieving a recall of 0.8748, accuracy of 0.92, and precision of 0.9394, compared to its performance with random weights. Similarly, VGG16 exhibits enhanced recall and precision with random weights, achieving a recall of 0.8862 and precision of 0.9005, while the model with ImageNet weights shows higher precision at 0.9865. Xception also benefits from ImageNet weights, particularly in recall and precision, with values reaching 0.9092 and 0.9618, respectively. Conversely, DenseNet121 shows consistent performance across both weight initialization methods, with recall values around 0.81 and precision above 0.93. Inception demonstrates improvements with ImageNet weights, particularly in accuracy and precision, achieving values of 0.9964 and 0.7922, respectively. VGG19, while showing perfect accuracy with random weights, achieves higher recall and precision with ImageNet weights of 0.7152 and 0.7904, respectively.

In particular, the proposed CNN model outperforms all others in terms of recall, achieving a value of 0.9871, and demonstrates perfect accuracy in all weight initialization scenarios.

The proposed model outperforms by achieving an impressive accuracy with a primary focus on distinguishing between two classes: 'DR' and 'No_DR.' As illustrated in Fig. 7, the proposed model demonstrated accurate predictions for 'DR' and 'No_DR'. Utilizing the same set of 550 test samples, the proposed model achieved a true positive rate of 266 for 'DR' and 265 for 'No_DR', with minimal errors.

Fig. 8 shows the accuracy and performance of the proposed model and how accurately it detects 'DR' and 'No_DR.' This compelling performance underscores the superiority of the proposed model over pre-trained transfer learning, machine learning, and deep learning models.

C. DIFFERENT DATASET PERFORMANCE EVALUATION

To address this issue, we employed two distinct datasets to assess the performance of the proposed model: Messidor2 and IDRiD [50]. The Messidor2 dataset was sourced from Kaggle at the following link: https://www.kaggle.com/datasets/mohammadasimbluemoon/diabeticretinopathy-messidor-eyepac-preprocessed, and the IDRiD dataset was obtained from https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid. The proposed model demonstrated an accuracy of 93.86% on the Messidor2 dataset and 91.18% on the IDRiD dataset. Table 7 shows a comprehensive insight into the robust performance of the proposed model across diverse datasets.

D. PERFORMANCE COMPARISON WITH EXISTING LITERATURE

A performance comparison is carried out with existing literature to show the efficacy of the proposed model. Table 8 provides a side-by-side comparison of the proposed model

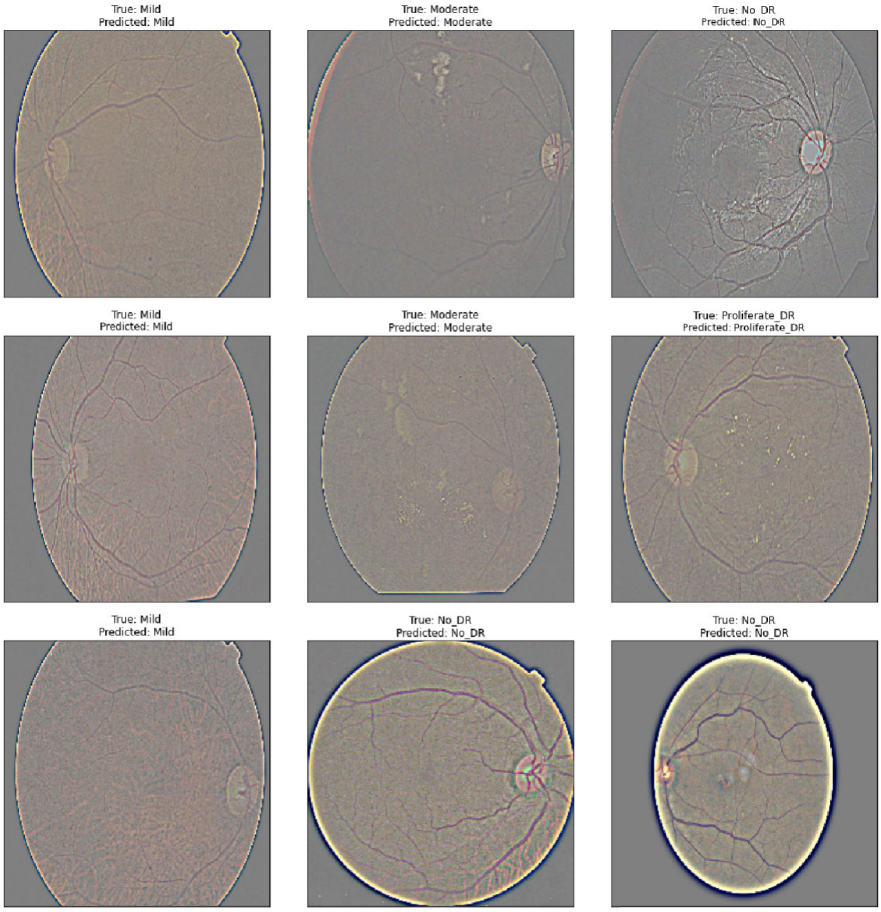


FIGURE 6. Multi-class detection using the proposed model.

TABLE 6. Comparison of transfer learning models trained for binary classification.

Model	Attributes	Trainable Parameters	Recall	Accuracy	Precision
ResNet50	Random weights	23544837	0.5406	1	0.9871
	Imagenet weights	23544837	0.8748	0.92	0.9394
VGG16	Random weights	14717253	0.8862	1	0.9005
	Imagenet weights	14717253	0.5802	1	0.9865
Xception	Random weights	20817197	0.8668	1	0.7458
	Imagenet weights	20817197	0.9092	0.7873	0.9618
DenseNet121	Random weights	6958981	0.8377	1	0.9314
	Imagenet weights	6958981	0.8139	0.7236	0.9819
Inception	Random weights	21778597	0.8607	1	0.9394
	Imagenet weights	21778597	0.8942	0.9964	0.7922
VGG19	Random weights	20026949	0.485	1	1
	Imagenet weights	20026949	0.7152	1	0.7904
CNN	Proposed	103104901	0.9871	1	0.9755

with existing approaches. Also, it shows how novel our approach was. Upon comparative analysis with existing research, it was observed that none of the prior studies specifically addressed multi-class detection; the prevailing focus has predominantly been on binary classification tasks. Furthermore, a notable absence of the incorporation of

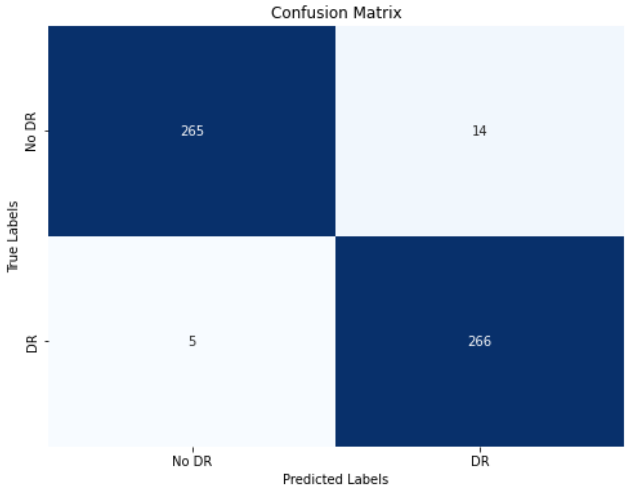


FIGURE 7. Confusion matrix of the proposed model.

TABLE 7. Model performance across different datasets.

Class Labels	Messidor2	IDRiD
No DR	1017	134
Mild	270	20
Moderate	347	136
Proliferate DR	75	74
Severe	35	49
Accuracy	93.86%	91.18%

XAI techniques was identified in the reviewed literature. In our contribution, we introduce novelty by integrating XAI methodologies. Our approach extends beyond binary classification, encompassing both binary and multi-class scenarios. Remarkably, the achieved accuracies surpassed those of existing works, underscoring the efficacy of our distinctive approach.

E. MODEL CHOOSE

The proposed CNN was selected from competing models (ResNet50, VGG16, Xception, DenseNet121, Inception, VGG19, RNN, KNN, SVM, and another CNN) based on its comprehensive architecture. Featuring five Conv2D layers, max-pooling, dropout regularization (dropout rate of 0.5), batch normalization, and dense layers culminating in a softmax activation function, the model excels in feature extraction, dimensionality reduction, and regularization. It achieved an outstanding recall score of 0.9585, accuracy of 0.9527, and precision of 0.9612 for multi-class classification and also achieved a recall score of 0.9871, accuracy of 1, and precision of 0.9755 for binary classification and a higher parameter count (103,104,901) shows superior performance in both multi-class and binary classifications. Notably, in both multi-class and binary classifications, the model achieves remarkable results, thus the model's robustness justifies its complexity, solidifying its position as the optimal choice for the specific classification task at hand.

TABLE 8. Summary of literature review and performance comparison with the proposed model.

Name	Proposed Method	Accuracy (%)
[15]	AlexNet CNN	92.50
[13]	KOA Based on BILSTM Network	84.09
[51]	Deep learning-based method	90.60
[52]	Hybrid deep learning	93.84
[53]	Linear Model and Naïve Bayes	75.13
[54]	Ensemble learning method	89.50
[55]	SVM and DL	84.60
[56]	Neural network	90.00
[22]	RetNet-10 method	98.65
[23]	DR-CCTNet method	90.17
Proposed	CNN(Proposed)	95.27

F. EXPALINABLE AI AND GRAD-CAM

The technique XAI [57] aims to offer insights into the reasoning behind AI systems' specific predictions or decisions, shedding light on the underlying logic, factors, and elements contributing to these choices. XAI methods can illuminate the inner workings of intricate deep learning models, enhancing their interpretability. Additionally, deep learning models are susceptible to biases inherent in the training data, which XAI can identify and mitigate by unveiling the most influential features or patterns in the predictions.

Grad-CAM [7] is an XAI technique utilized for visualizing and elucidating the decision-making process of CNNs in tasks related to image classification. The primary function of Grad-CAM is to identify the regions or features within an input image that exert the most influence on the CNN's prediction for a specific class. Grad-CAM proves particularly beneficial for unraveling the reasoning behind the CNN's individual predictions, thereby offering insights into the intricate process guiding the model's decision-making. In addition to the statistical metrics conducive to research, Grad-CAM's output also facilitates qualitative model analysis [58].

Fig. 9 shows the Grad-CAM results generated by the proposed model using random images from the test set. In the depicted image, 'True 1' and 'pred 1' denote positive results, indicating the presence of DR, while 'True 0' and 'pred 0' signify negative results, suggesting the absence of DR. Conversely, if 'True 1' and 'pred 0' or 'True 0' and 'pred 1' are observed, it implies suboptimal performance by Grad-CAM. In our case, we initially obtain visual output from Fig. 6 and Fig. 8, following which we apply Grad-CAM to cross-validate the results for better comprehension by humans and medical professionals. Moreover, our Grad-CAM results demonstrate accurate DR and No DR diagnosis by highlighting specific areas. In Fig. 9, where DR is present, it is distinctly highlighted, corresponding to 'True 1' and 'pred 1,' indicating the presence of DR. Conversely, the 'True 0' and 'pred 0' regions signify the absence of DR. Ultimately,

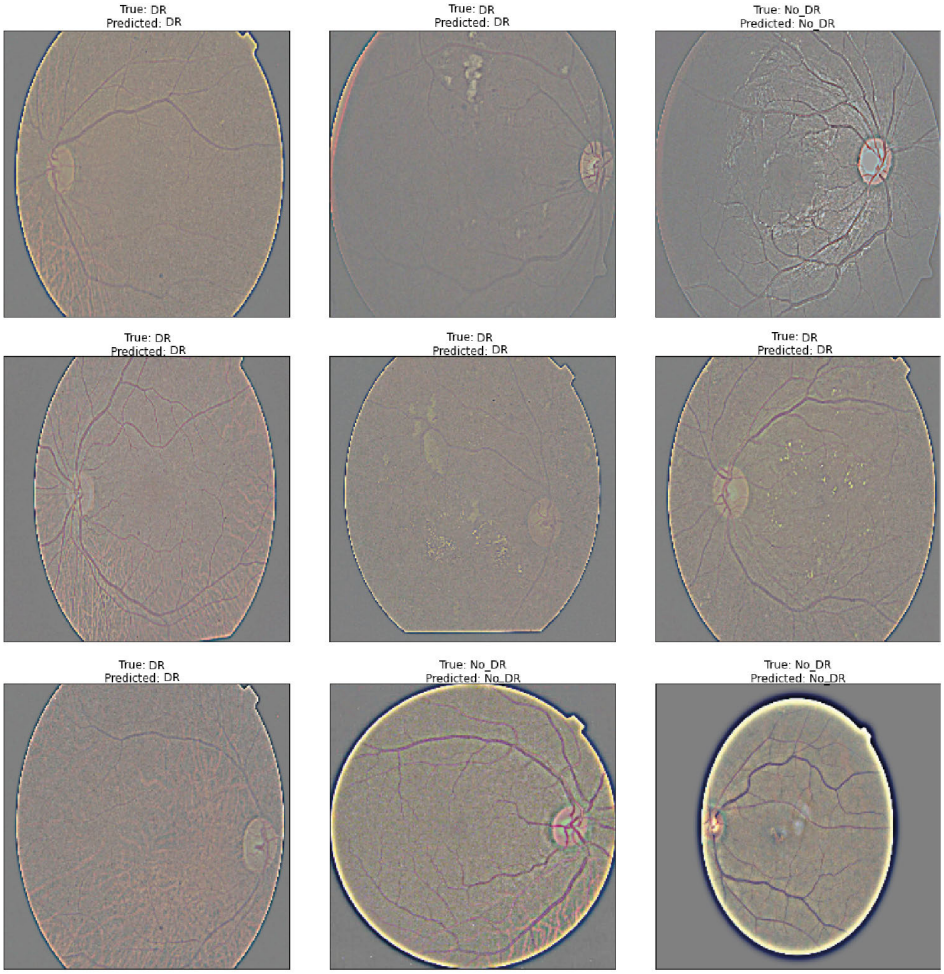


FIGURE 8. Binary class detection using the proposed model.

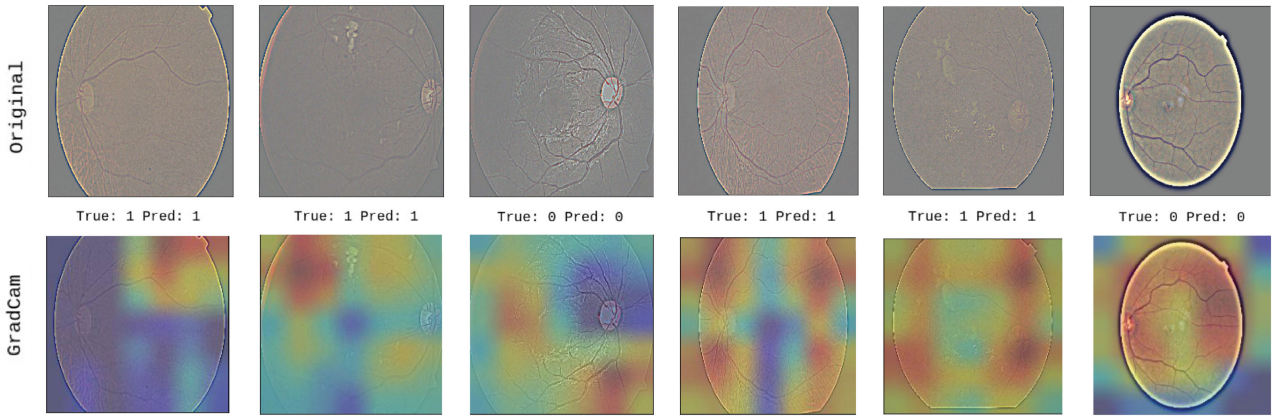


FIGURE 9. Grad-cam results from the test set.

Grad-CAM enriches presentations by elucidating the CNN model's decisions, making them more comprehensible to a wider audience.

V. CONCLUSION

The goal of this study is to develop a model for the early diagnosis of diabetic retinopathy that can classify images based on the disease's development. In this study, we have used high-resolution color fundus retinal pictures. Since a wide range of fundus images were present, there were some imbalances and challenges present as well. However, a variety of image modification and data augmentation techniques have been used in this study: Image flipping (Horizontal, Vertical and Random Rotation), Image resampling, Image zooming and Constant Filling. Additionally, the number of images has been increased so that upon augmentation, every

label image would have almost the same length. As a result, the dataset had 28,059 images overall and went through several cycles to fine-tune the pre-processing phases to produce the greatest possible image quality. The dataset was augmented and several pre-processing techniques have also been applied. In this work, 1928 pictures were set aside for validation and a total of 28059 photos were used for training. Transfer learning models such as DenseNet121, Xception, Resnet50, VGG16, VGG19, and InceptionV3, machine learning models such as SVM, and neural network models like RNN have been employed for binary and multi-class classification followed by a comprehensive comparison among them. Two different image sizes have been used for this study, for the 32 × 32 image size, the average time per epoch was roughly 20 seconds, and for the 16 × 16 image size, the average time per epoch was only 10 seconds. Xception performed best among the transfer learning models. However, our proposed CNN model outperformed among all the other models with an accuracy of 95.27% accuracy for multi-class detection in all types of DR (Mild, Moderate, Proliferate DR and Severe) and No_DR diagnosis, and with a perfect accuracy of 1 for binary classification, incorporating XAI and Grad-CAM. To show how well our new model works, we evaluated the model on different datasets as well where the proposed model demonstrated an accuracy of 93.86% on the Messidor2 dataset and 91.18% on the IDRiD dataset. Moreover, for fair comparison, we compared it to other research and it was found that none of the previous studies had a special focus on multi-class detection with greater accuracy incorporating Grad-Cam. Additionally, we offer a comprehensive comparison of various models to identify the optimal detection model, integrating XAI and Grad-CAM, along with the addition of different optimizers to enhance performance. Our future plan is to develop a well-balanced dataset, particularly focusing on regions with a high prevalence of diabetes like in Bangladesh. Moreover, we aim to create a platform allowing users to upload images for real-time output, facilitating immediate decision-making for this disease. Furthermore, we intend to develop a more suitable and accurate model for enhanced directionality. The goal of this study was to help doctors identify severity levels precisely for better treatment decisions by classifying each stage of diabetic retinopathy and the findings of this study will benefit researchers and healthcare professionals, inspiring further research to enhance disease detection in medical imaging on a global scale.

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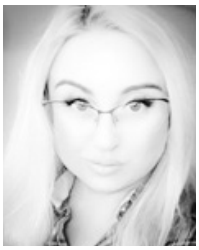


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